

STA322 - Project 1

Julia Zheng and Christina Zhang

Introduction

```
library(survey)
```

```
Loading required package: grid
```

```
Loading required package: Matrix
```

```
Loading required package: survival
```

```
Attaching package: 'survey'
```

```
The following object is masked from 'package:graphics':
```

```
dotchart
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
x tidyr::expand() masks Matrix::expand()
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
x tidyr::pack() masks Matrix::pack()
x tidyr::unpack() masks Matrix::unpack()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(pps)
library(dplyr)
```

```
# load and reorganize data
```

```
trinity <- read_csv("trinity_faculty.csv")
```

```
Rows: 41 Columns: 5
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Dept name
```

```
dbl (4): Stratum #, # of faculty, Proportional allocation, Rounded allocation
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trinity <- trinity |>
  select(department = "Dept name", M_i = "# of faculty")

# certainty departments (pi_i = 1), same 7 as in the sampling step
certainty_depts <- c("Economics", "Physics", "Romance Studies", "Music",
  "Biology", "Mathematics", "Psychology & Neuroscience")

certainty <- trinity |>
  filter(department %in% certainty_depts) |>
  mutate(pi_i = 1)

# remaining departments: 13 more selected, pi_i = 13 * M_i / sum(M_i)
remaining <- trinity |>
  filter(!department %in% certainty_depts)
remaining$pi_i <- 13 * remaining$M_i / sum(remaining$M_i)

max(remaining$pi_i) # should be 0.9246, matching the sampling step
```

```
[1] 0.9246088
```

```
dept_info <- bind_rows(certainty, remaining)
```

```
# systematic PPS sampling
```

```
n_dept <- 20
```

```
t_x <- sum(trinity$M_i) # total number of faculty across all departments
```

```
trinity$pi_i <- n_dept * trinity$M_i / t_x # first-order inclusion probability for each department
```

```
max(trinity$pi_i) # check assumption that  $n_{xi}/t_x \leq 1$  for all  $i$ . it's not...
```

```
[1] 1.378151
```

```
# check which department p_i > 1 --> treat as certainty cases
```

```
trinity |>
```

```
  filter(pi_i > 1) |>
```

```
  select(department, M_i, pi_i)
```

```
# A tibble: 6 x 3
```

	department	M_i	pi_i
	<chr>	<dbl>	<dbl>
1	Biology	77	1.29
2	Economics	81	1.36
3	Mathematics	82	1.38
4	Music	62	1.04
5	Physics	60	1.01
6	Romance Studies	66	1.11

```
# dealing with certainty cases
```

```
# separate initial certainty and non-certainty departments
```

```
certainty <- trinity |>
```

```
  filter(pi_i >= 1) |>
```

```
  mutate(pi_i = 1)
```

```
remaining <- trinity |>
```

```
  filter(pi_i < 1)
```

```

n_remaining <- n_dept - nrow(certainty) # calculate number of additional departments to select
remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i) # recalculate pi_i for remaining
# if recalculation creates additional certainty PSUs, repeat until all remaining probabilities are 1
while (any(remaining$pi_i >= 1)) {
  new_certainty <- remaining |>
    filter(pi_i >= 1) |>
    mutate(pi_i = 1)
  certainty <- bind_rows(certainty, new_certainty)
  remaining <- remaining |>
    filter(pi_i < 1)
  n_remaining <- n_dept - nrow(certainty)
  remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i)
}

# results check
certainty |>
  select(department, M_i, pi_i) # 7 certainty cases identified (1 additional throughout recalculation)

```

```

# A tibble: 7 x 3
  department      M_i pi_i
  <chr>      <dbl> <dbl>
1 Biology      77     1
2 Economics    81     1
3 Mathematics  82     1
4 Music        62     1
5 Physics      60     1
6 Romance Studies 66     1
7 Psychology & Neuroscience 59     1

```

```
nrow(certainty)
```

```
[1] 7
```

```
n_remaining
```

```
[1] 13
```

```
max(remaining$pi_i)
```

```
[1] 0.9246088
```

```
set.seed(919)
```

```
# select remaining cases with pps & add certainty cases
```

```
selected_depts <- ppss(remaining$M_i, n_remaining) # returns numbers of selected rows
```

```
pps_remaining <- remaining[selected_depts, ] # keep selected departments
```

```
pps_sample <- bind_rows(certainty, pps_remaining) # combine certainty psus with pps-selected
```

```
nrow(pps_sample) # check 20 departments were selected
```

```
[1] 20
```

```
# view selected departments
```

```
pps_sample |>
```

```
  select(department, M_i, pi_i)
```

```
# A tibble: 20 x 3
```

	department	M_i	pi_i
	<chr>	<dbl>	<dbl>
1	Biology	77	1
2	Economics	81	1
3	Mathematics	82	1
4	Music	62	1
5	Physics	60	1
6	Romance Studies	66	1
7	Psychology & Neuroscience	59	1
8	Art, Art History & Visual Studies	44	0.814
9	Asian & Middle Eastern Studies	45	0.832
10	Cinematic Arts	7	0.129
11	Computer Science	48	0.888
12	Dance Program	21	0.388
13	English	50	0.925
14	Gender, Sexuality & Feminist Studies	13	0.240
15	History	47	0.869
16	Philosophy	24	0.444

17 Political Science	43 0.795
18 Sociology	35 0.647
19 Statistical Science	32 0.592
20 Writing and Rhetoric Program	31 0.573

```
# stage 2 srs

# srs 5 faculty from each department
pps_sample$faculty_numbers <- lapply(
  pps_sample$M_i,
  function(M_i) sample(1:M_i, 5)
)

# view numbers of faculty selected
pps_sample |>
  select(department, M_i, pi_i, faculty_numbers)
```

```
# A tibble: 20 x 4
  department M_i pi_i faculty_numbers
  <chr>      <dbl> <dbl> <list>
1 Biology    77 1    <int [5]>
2 Economics  81 1    <int [5]>
3 Mathematics 82 1    <int [5]>
4 Music      62 1    <int [5]>
5 Physics    60 1    <int [5]>
6 Romance Studies 66 1    <int [5]>
7 Psychology & Neuroscience 59 1    <int [5]>
8 Art, Art History & Visual Studies 44 0.814 <int [5]>
9 Asian & Middle Eastern Studies 45 0.832 <int [5]>
10 Cinematic Arts 7 0.129 <int [5]>
11 Computer Science 48 0.888 <int [5]>
12 Dance Program 21 0.388 <int [5]>
13 English   50 0.925 <int [5]>
14 Gender, Sexuality & Feminist Studies 13 0.240 <int [5]>
15 History   47 0.869 <int [5]>
16 Philosophy 24 0.444 <int [5]>
17 Political Science 43 0.795 <int [5]>
18 Sociology  35 0.647 <int [5]>
19 Statistical Science 32 0.592 <int [5]>
20 Writing and Rhetoric Program 31 0.573 <int [5]>
```

```

samples <- read.csv("pps_sample.csv")

samples <- samples |>
  select(-any_of(c("M_i", "pi_i", "pi", "weight"))) |>
  rename(
    woman = Sex..1...F..0...other.,
    grant = Ongoing.grant...1...Yes..0...No.,
    has_phd = PhD...1...Yes..0...No.,
    yrs_phd = Number.of.years.since.Phd,
    yrs_pub = Years.since.last.pub,
    intl_ugrad = International.undergrad.degree.
  )

# 3. attach M_i and pi_i, then build the weight
samples <- samples |>
  left_join(dept_info, by = "department")

samples$weight <- (1 / samples$pi_i) * (samples$M_i / 5)

# 4. survey design
fac_des <- svydesign(
  ids = ~department,
  weights = ~weight,
  data = samples
)

```

Question 1. What is an estimate of the fraction of faculty who are women?

```

#|label: women-prop

prop_women <- svymean(~woman, fac_des)
prop_women

```

```

      mean    SE
woman 0.35968 0.062

```

```

confint(prop_women, level = .95)

```

```

      2.5 %    97.5 %
woman 0.2381951 0.4811584

```

Question 2. What fraction of faculty have at least one grant that is ongoing in 2026?

```
#|label: grant
```

```
prop_grant <- svymean(~grant, fac_des)
prop_grant
```

```
      mean      SE
grant 0.18953 0.0539
```

```
confint(prop_grant, level = .95)
```

```
      2.5 %    97.5 %
grant 0.08396691 0.2950893
```

Question 3. Among faculty who have a Ph.D. degree, what is the average number of years since they received the Ph.D. degree?

```
#|label: phd
```

```
phd_des <- subset(fac_des, has_phd == 1)

avg_phd <- svymean(~yrs_phd, phd_des)
avg_phd
```

```
      mean      SE
yrs_phd 25.665 2.4335
```

```
confint(avg_phd, level = .95)
```

```
      2.5 %    97.5 %
yrs_phd 20.89577 30.43481
```

Question 4.

a) Among those who have published papers, what is the avg number of years since their last publication?

```
#|label: pub-yrs

pub_des <- subset(fac_des, !is.na(yrs_pub))

avg_pub <- svymean(~yrs_pub, pub_des)
avg_pub
```

```
      mean      SE
yrs_pub 2.8581 0.6312
```

```
confint(avg_pub, level = .95)
```

```
      2.5 %   97.5 %
yrs_pub 1.620912 4.095311
```

b) Did this faculty receive their undergraduate degree outside of the US?

```
#|label: intl-ug

prop_intl <- svymean(~intl_ugrad, fac_des)
prop_intl
```

```
      mean  SE
intl_ugrad  NA NaN
```

```
confint(prop_intl, level = .95)
```

```
      2.5 % 97.5 %
intl_ugrad  NaN   NaN
```